

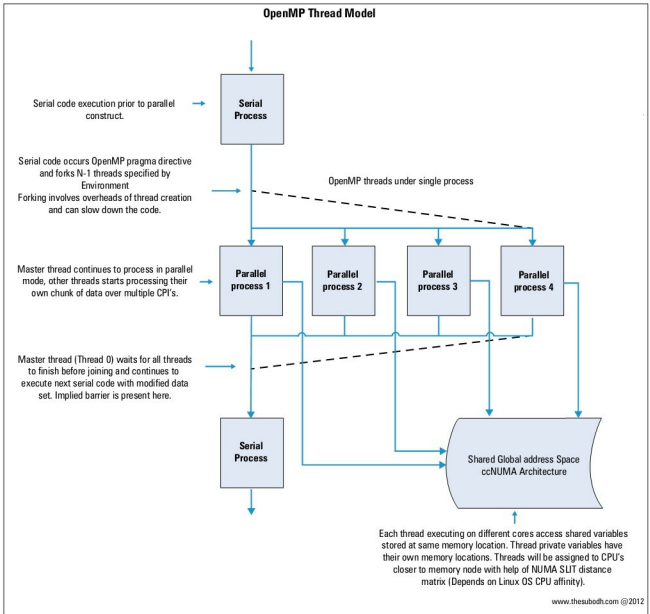


Figure 1: OpenMP Thread Model

cause a decrease in performance by stealing CPU time, though this behaviour depends upon the application and system configurations.

Advantages of OpenMP

- Easy to program. More readable code.
- Data layout is handled automatically by OpenMP directives.
- Threads can move from one core to another, unless specifically bound.
- Easy modification of environment settings.
- Guaranteed to perform efficiently on Shared Memory Systems.
- Portable and widely available.

Disadvantages

- Performance may decrease after a finite number of threads, as all threads will compete for shared memory bandwidth. To reduce this, threads must be placed properly closer to CPUs, depending upon the application's requirements.
- Thread synchronisation is expensive.
- Available on SMP (Symmetric Multi-processing) systems only.
- Not as scalable as MPI (Message Passing Interface), although a hybrid model of MPI + OpenMP + OpenMP CUDA is getting a lot of attention.

Parallel matrix multiplication

As part of learning OpenMP, I have written code for Parallel Matrix Multiplication. The algorithm used is a